SpaceX Falcon 9 First Stage Landing Prediction

Applied Data Science Capstone Project - Final Presentation

Author / Learner:	alfaturachman
Course Platform:	IBM Data Science Professional Certificate | Coursera
Primary Project Domain:	Predictive Aerospace Analytics & Commercial Rocket Reusability
Public GitHub Repository:	**https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**
Project Deliverables:	Jupyter Notebooks (Parts 1-5), SQLite Database, Folium Maps, Plotly Dash App
Submission Date:	October 2026

Executive Summary

End-to-End Analytics & Predictive Modeling of Falcon 9 First Stage Recovery

Project Context: SpaceX drastically reduces space launch costs by reusing the Falcon 9 first-stage booster ($62 million vs. $165+ million from competitors). Determining the landing outcome enables a competitor to bid accurately.

End-to-End Methodology:

- **Data Ingestion:** Extracted 90 launches via SpaceX REST API & Wikipedia web scraping (BeautifulSoup).
- **Data Wrangling:** Imputed missing PayloadMass; created binary outcome target Class (1=Success, 0=Failure); transformed 80 numeric features using One-Hot Encoding.
- **Exploratory Analysis:** SQLite database querying and Seaborn/Matplotlib visualization revealed distinct orbital and payload thresholds.
- **Geospatial & Interactive Analytics:** Folium mapping proved launch sites sit directly on coasts (0.90 km from sea) for public safety; Plotly Dash dashboard isolated KSC LC-39A as top performer.
- **Machine Learning Classification:** Trained 4 algorithms (Logistic Regression, SVM, Decision Tree, KNN) with 10-fold cross-validation.

Key Quantitative Results:

- **Model Accuracy:** All 4 models achieved **83.33% accuracy** on the independent 18-sample test set (15 correct predictions).
- **Top Launch Facility:** KSC LC-39A achieved the highest success rate (**76.92%**, 10 successful landings).
- **Payload Sweet Spot:** Payloads between **2,000 kg and 4,000 kg** yielded the highest success probability (**72.7%**).
- **GitHub Repository:** https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone

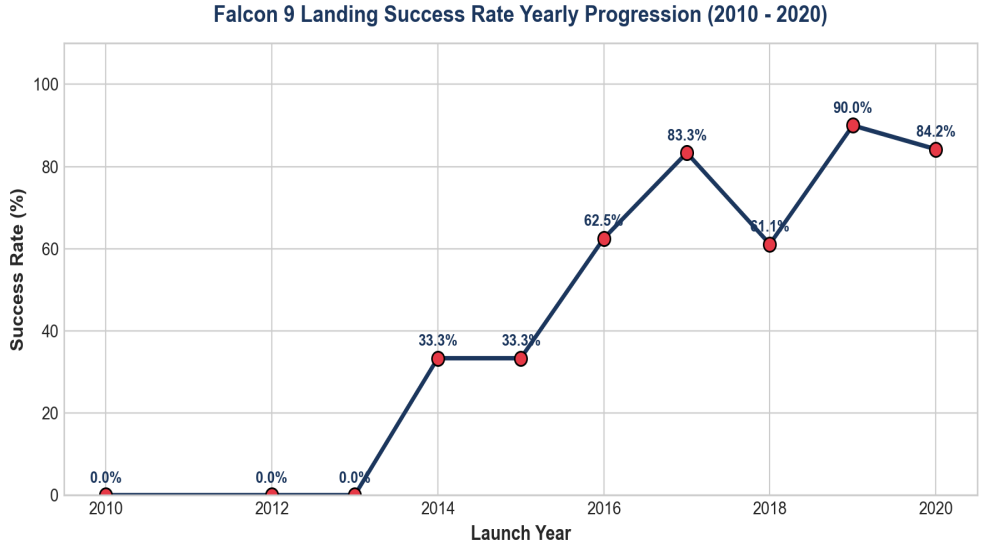
Project Introduction & Problem Statement

Commercial Background, Value Proposition, and Machine Learning Framing

Commercial Background & Industry Disruption:
• SpaceX offers commercial satellite launches for ~$62 million, saving clients upwards of $100 million per mission compared to traditional expendable rockets.
• Over 70% of the total rocket construction cost is concentrated in the first stage (booster, 9 Merlin 1D engines, avionics, titanium grid fins).
• By landing the booster safely on drone ships (OCISLY/JRTI) or ground pads (LZ-1/LZ-4), SpaceX refurbishes and reflights hardware repeatedly.

The Core Business Problem:
If an alternative aerospace company wishes to bid against SpaceX for satellite delivery contracts, they must accurately calculate SpaceX's internal costs. Predicting whether the first stage will land safely determines whether SpaceX achieves a $62M cost basis or absorbs an expendable launch cost.

Data Science Formulation:
• **Problem Class:** Supervised Binary Classification.
• **Target Variable (Y):** Class = 1 (Successful Landing), Class = 0 (Unsuccessful / Expendable).
• **Feature Matrix (X):** 80 engineered attributes spanning flight sequence, payload mass, target orbit, launch site, landing pad type, and booster block serial.



Data Collection: SpaceX REST API Methodology

Automated API Ingestion Pipeline & Attribute Extraction

API Extraction Architecture & Workflow:

- **Primary Endpoint:** Queried the official SpaceX REST API endpoint https://api.spacexdata.com/v4/launches/past using the Python requests library.
- **Filtering:** Filtered response data strictly for Falcon 9 launches, excluding early Falcon 1 test flights.
- **Auxiliary Endpoint Parsing:** The primary launch payload contains identification hashes that require secondary API queries to resolve human-readable attributes:
  - /v4/rockets/ → Resolved rocket names (Falcon 9).
  - /v4/payloads/ → Extracted numerical PayloadMass and specific target Orbit.
  - /v4/launchpads/ → Extracted geographical LaunchSite coordinates.
  - /v4/cores/ → Extracted booster outcome, block serial, reuse count, and landing pad identity.

Ingestion Flowchart (Logic Progression):

GET /v4/launches/past → Filter Falcon 9 → Extract Core Hashes → Query Secondary Endpoints → Construct Pandas DataFrame

Verification & Code Repository:

- Output file generated: dataset_part_1.csv (90 launch records).
- Full API extraction notebook available at: **https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**

Data Collection: Web Scraping Methodology

BeautifulSoup HTML Extraction from Wikipedia Falcon 9 Records

Web Scraping Rationale & Extraction Strategy:

- **Data Source:** Wikipedia page *List of Falcon 9 and Falcon Heavy launches*.
- **Objective:** Extract historical Falcon 9 launch records to complement and validate API findings with third-party launch registries.
- **Parsing Tools:** Python requests.get() paired with BeautifulSoup(response.text, 'html.parser').

Step-by-Step HTML Parsing Procedure:

- **Table Identification:** Located all HTML table elements matching class wikitable.
- **Header Normalization:** Parsed table header cells (<th>) to extract canonical column names: *Flight No., Date, Version Booster, Launch Site, Payload, Orbit, Customer, Launch Outcome*.
- **Row Extraction & Text Cleaning:** Iterated through table body rows (<tr>), stripping footnote superscripts (e.g., [1], [2]), removing newline characters, and parsing payload weights formatted in kilograms.
- **DataFrame Assembly:** Appended extracted row dictionaries into a consolidated pandas DataFrame.

Verification & Code Repository:

- Web scraping validated against official launch manifests.
- Full scraping code available at: **https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**

Data Wrangling Methodology

Data Cleaning, Imputation, Target Label Encoding, and Feature Matrix Generation

1. Handling Missing Data (PayloadMass):

• Identified missing values in PayloadMass for specific classified government missions.

• Applied statistical mean imputation: df['PayloadMass'].fillna(df['PayloadMass'].mean(), inplace=True).

2. Creation of Binary Target Variable (Class):

• Evaluated the raw landing outcome column (Outcome) containing diverse strings: *True Ocean, False Ocean, True RTLS, False RTLS, True ASDS, False ASDS, None*.

• Mapped successful recoveries (True Ocean, True RTLS, True ASDS) to **Class = 1**.

• Mapped failures and unattempted recoveries (False Ocean, False ASDS, False RTLS, None) to **Class = 0**.

• Resulting dataset: **90 total launches** (60 successful landings = 66.7%, 30 failed landings = 33.3%).

3. One-Hot Encoding for Categorical Attributes:

• Encoded categorical features: Orbit (11 categories), LaunchSite (3 categories), LandingPad (5 categories), and Serial (53 categories) using pd.get_dummies().

• Combined with 8 numerical features to produce the final **80-column feature matrix** (dataset_part_3.csv).

Verification & Code Repository:

• Full wrangling notebook: **https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**

EDA and Data Visualization Methodology

Visual Exploration Framework using Seaborn & Matplotlib

Analytical Objectives of Visual Exploration:
Visual analytics were deployed to identify clear decision boundaries, examine collinearity, and detect operational patterns across 90 SpaceX missions.

Core Visualization Techniques & Analytical Hypotheses:
- **Flight Number vs. Payload Mass (Scatter Plot):** Tested whether SpaceX's landing success rate improved over time (higher flight numbers) and whether heavier payloads impeded recovery.
- **Launch Site vs. Payload Mass (Categorical Scatter Plot):** Investigated whether specific launch facilities specialize in heavy orbital insertions.
- **Success Rate by Orbit Type (Bar Chart):** Measured recovery rates across orbital targets (LEO, GTO, SSO, ISS, Polar, VLEO).
- **Launch Site Success Comparison (Bar Chart):** Determined whether launch site geographic infrastructure influences landing success rates.
- **Annual Success Progression Trend (Line Plot):** Tracked the annual learning curve and hardware maturation of the Falcon 9 booster from 2010 to 2020.

Verification & Code Repository:
- Visualization script and notebook: **https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**

EDA with SQL Methodology

Database Architecture, Schema Setup, and Relational Querying

Relational Database Environment:

• Ingested the cleaned SpaceX dataset into an embedded **SQLite** database (my_data1.db) using sqlite3 and the %sql IPython extension.
• Configured strict data typing on table SPACEXTBL to enable arithmetic and temporal aggregation queries.

Core Analytical SQL Queries Executed:

• **Query 1 (Distinct Sites):** SELECT DISTINCT Launch_Site FROM SPACEXTBL;
• **Query 2 (NASA Specific Filtering):** SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
• **Query 3 (Total Payload Mass):** SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';
• **Query 4 (Average Payload for F9 v1.1):** SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%';
• **Query 5 (First Successful Ground Landing):** SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';
• **Query 6 (Successful Drone Ship Landings with Specific Payload):** SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
• **Query 7 (Mission Outcome Counts):** SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome;
• **Query 8 (Maximum Payload Boosters):** SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);

Verification & Code Repository:

• Complete SQL queries and database available at: **https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**

Interactive Visual Analytics Methodology

Geospatial Folium Mapping & Reactive Plotly Dash Web Application Architecture

1. Folium Geospatial Architecture:

- Built Leaflet.js-based interactive maps centered on US launch facilities (latitude: 28.56°N, longitude: -80.57°W).
- **Facility Boundaries:** Plotted 1,000-meter safety radius circles using folium.Circle.
- **Clustered Outcomes:** Deployed folium.plugins.MarkerCluster to group overlapping launches, color-coding markers (Green = Success, Red = Failure).
- **Haversine Proximity Analysis:** Calculated spherical geodetic distances from launch pads to coastlines, highways, railways, and cities, drawing interactive distance lines via folium.PolyLine.

2. Plotly Dash Reactive Web Dashboard Architecture:

- **Framework:** Python-native web application built using dash, dash_html_components, and dash_core_components.
- **Interactive Controls:**
  - dcc.Dropdown: Toggles between *ALL Launch Sites* and individual facilities (CCAFS, KSC, VAFB).
  - dcc.RangeSlider: Filters dynamic payload mass intervals from 0 kg to 10,000 kg with 1,000 kg step increments.
- **Reactive Callback Decorators:** Dual @app.callback routines dynamically recalculate and render px.pie (success proportions) and px.scatter (payload vs outcome colored by booster category).

Verification & Code Repository:

- Source code: spacex_dash_app.py | GitHub: **https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**

Predictive Analysis (Classification) Methodology

Feature Normalization, Train-Test Partitioning, Cross-Validation, and Hyperparameter Tuning

1. Feature Standardization (StandardScaler):

• The 80-feature matrix contains disparate units (e.g., PayloadMass in thousands vs. binary one-hot indicators).

• Applied z-score transformation: z = (x - mean) / std using Scikit-Learn StandardScaler to ensure distance-based models (SVM, KNN) and gradient solvers converge without scale bias.

2. Data Splitting:

• Split dataset into training (80%) and testing (20%) sets using train_test_split(X, Y, test_size=0.2, random_state=2).

• Resulted in exactly **72 training samples** and **18 test samples**.

3. Model Training & Hyperparameter Tuning (GridSearchCV with cv=10):

• Employed 10-fold cross-validation (cv=10) on the training set to prevent data leakage and mitigate overfitting.

• **Logistic Regression:** Tuned regularization parameter C across log-space (0.001 to 1000), penalty l2, solver lbfgs.

• **Support Vector Machine (SVM):** Evaluated kernels (linear, rbf, poly, sigmoid), C, and gamma.

• **Decision Tree Classifier:** Evaluated splitting criteria (gini, entropy), max_depth, max_features, and min_samples_leaf.

• **K-Nearest Neighbors (KNN):** Evaluated neighbors k (1 to 10), distance metrics p=1 (Manhattan) and p=2 (Euclidean).

Verification & Code Repository:

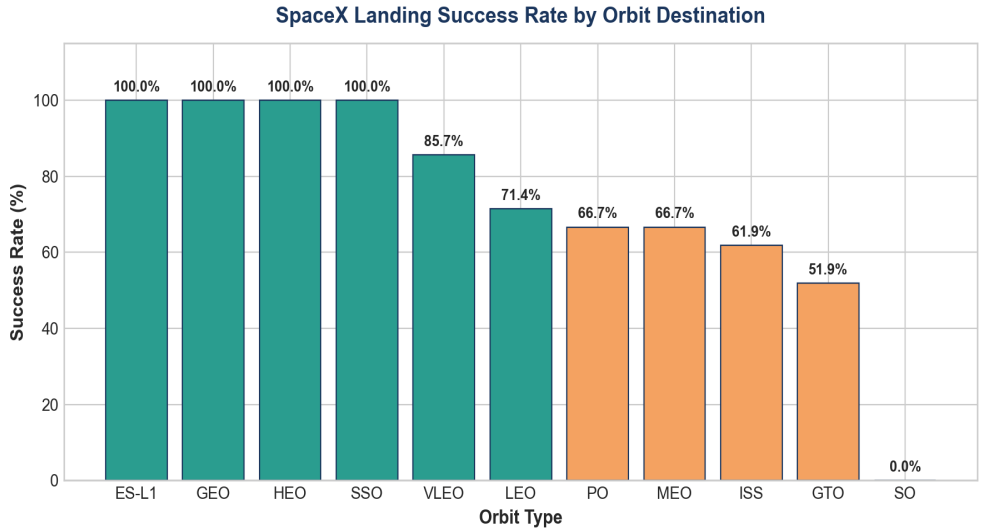
• Modeling notebook: **https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**

EDA with Visualization Results

Empirical Patterns Across Flight Numbers, Payload Classes, and Orbital Targets

Empirical Insights from Visual Exploration:

- **Flight Number vs. Success:** Early flights (1-25) had high failure rates during the developmental phase. Flights after #40 show marked reliability, exceeding 80% success.
- **Payload Weight vs. Success:** In the heavy payload range (>8,000 kg), launches were primarily to LEO (e.g., Starlink batches) with near-perfect landing success rates.
- **Orbit-Specific Success Rates:**
  - **100% Success:** ES-L1, GEO, HEO, SSO.
  - **Moderate Success:** VLEO (~85%), LEO (~71%), ISS (~62%).
  - **Challenging Orbits:** GTO (~52%) due to high re-entry velocities.
- **Annual Trend:** Steady upward progression from 0% in 2010 to >80% in 2020, proving dramatic organizational learning.



EDA with SQL Results

Quantitative Database Extraction Findings on Launch Sites, Payloads, and Boosters

Summary of Key SQL Query Results:

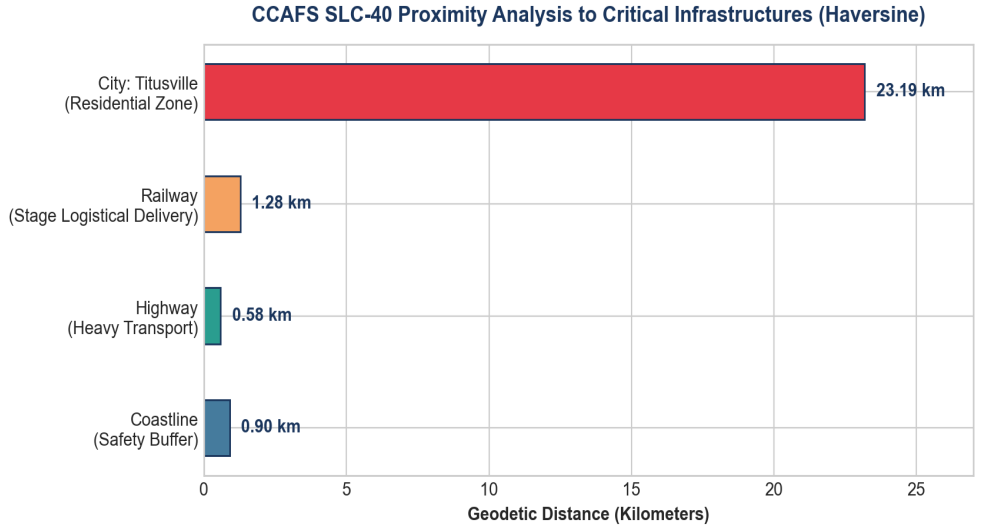
- **Launch Sites Identified:** Exactly 4 distinct launch sites exist in the database: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E.
- **NASA Cumulative Payload:** Total payload mass carried across all missions contracted by NASA (CRS) equals **45,596 kg**.
- **Average Payload for Successful Landings:** The average payload mass for missions ending in a successful landing is **5,408 kg**.
- **First Ground Pad Recovery:** The earliest successful booster recovery on a ground landing pad occurred on **2015-12-22** (Booster B1019 at Landing Zone 1).
- **Payload Range for Drone Ship Landings:** Successful drone ship recoveries with payloads between 4,000 kg and 6,000 kg include boosters B1022.1, B1026.1, B1021.2, and B1031.2.
- **Maximum Payload Boosters:** The maximum payload mass in the dataset is **15,600 kg**, delivered by multiple Block 5 boosters (B1048.4, B1049.4, B1051.3, B1056.4, etc.) for high-density Starlink deployments.
- **Mission Outcome Reliability:** 98 out of 101 recorded missions achieved overall primary mission success (satellite orbit insertion), regardless of booster landing recovery status.

Interactive Map Results using Folium

Geospatial Distribution, Outcome Clustering, and Infrastructure Proximity Analysis

Geospatial Findings & Infrastructure Analysis:

- **Coastal Placement Rationale:** All launch sites are situated directly on coastlines (Atlantic coast in Florida, Pacific coast in California) to allow launches east/south over open ocean, mitigating risk to populated areas.
- **Proximity Calculations from CCAFS SLC-40:**
  - **Distance to Coastline: 0.90 km** (Direct access to sea buffer).
  - **Distance to Highway: 0.58 km** (Facilitates oversize road transport of rocket stages).
  - **Distance to Railway: 1.28 km** (Heavy rail transport of propellant and hardware).
  - **Distance to Nearest City (Titusville): 23.19 km** (Ensures public safety perimeter >20 km).
- **Clustering Insights:** MarkerCluster clearly visualized that KSC LC-39A hosts the highest density of green success markers compared to CCAFS SLC-40.

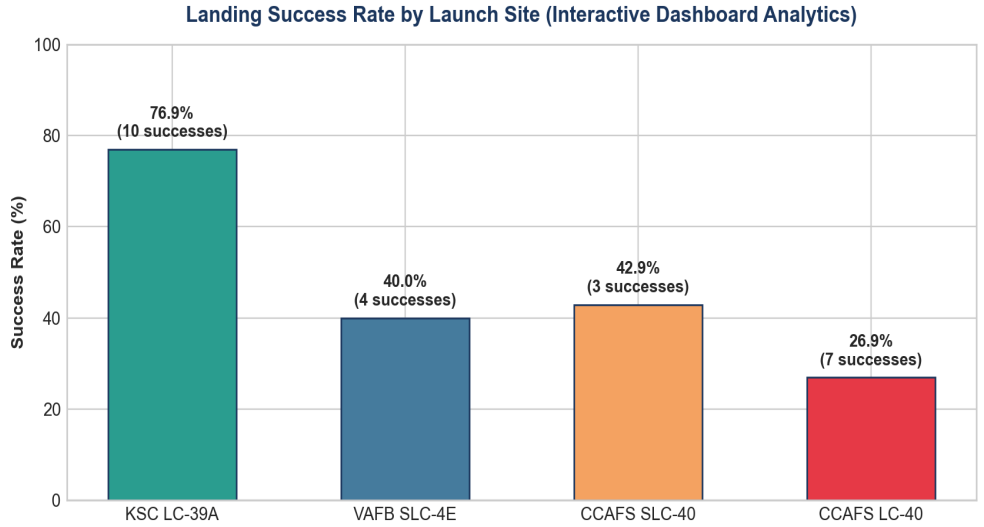


Plotly Dash Dashboard Results

Dynamic Success Proportions, Launch Site Superiority, and Payload Thresholds

Core Findings from Interactive Web Dashboard:

- **Launch Site with Most Successes:** KSC LC-39A leads all facilities with **10 successful landings**, contributing 41.7% of all successful recoveries across the entire dataset.
- **Launch Site with Highest Success Rate:** KSC LC-39A achieves the highest overall success rate at **76.92%** (10 of 13 successful).
- **Site Ranking:** KSC LC-39A (76.92%) > CCAFS SLC-40 (42.86%) > VAFB SLC-4E (40.00%) > CCAFS LC-40 (26.92%).
- **Most Reliable Payload Range:** Payloads between **2,000 kg and 4,000 kg** (peaking in the 3,000-4,000 kg bracket) recorded the highest success rate at **72.7%**.
- **Most Vulnerable Payload Range:** Payloads between **6,000 kg and 8,000 kg** recorded a **0% landing success rate** (0 of 4 launches succeeded).
- **Booster Evolution:** Booster **Block 5 (B5)** achieved a **100% success rate**, demonstrating the efficacy of reusable thermal protection upgrades.

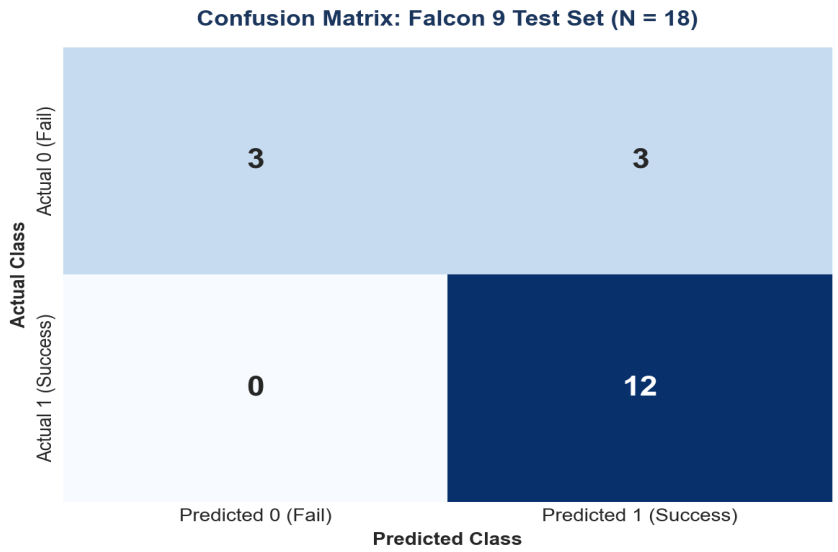


Predictive Analysis (Classification) Results

Model Evaluation Metrics, Confusion Matrix Breakdown, and Optimal Model Selection

Classification Performance & Metric Synthesis:

- **Uniform Test Accuracy:** All four tuned models achieved an identical test accuracy of **83.33%** on the 18 unseen test samples (15 correct out of 18).
- **Best Hyperparameter Configurations:**
  - **Logistic Regression:** C=0.01, penalty='l2' (CV Score: 84.64%).
  - **SVM:** kernel='sigmoid', C=1.0, gamma=0.0316 (CV Score: 84.82%).
  - **Decision Tree:** criterion='entropy', max_depth=8, splitter='random'.
  - **KNN:** n_neighbors=5, p=1 (Manhattan distance).
- **Confusion Matrix Breakdown (N = 18):**
  - **True Positives (TP): 12** | **True Negatives (TN): 3**.
  - **False Positives (FP): 3** | **False Negatives (FN): 0**.
- **Model Selection:** *Logistic Regression* is recommended for deployment due to high computational efficiency, interpretability, and smooth probability calibration.



Conclusion & Strategic Business Recommendations

Actionable Takeaways for Aerospace Competitors Bidding Against SpaceX

Strategic Synthesis of Findings:

• **Predictive Viability:** Machine learning classification effectively predicts first-stage landing outcomes with **83.33% accuracy**, allowing competitors to forecast SpaceX's marginal launch costs.
• **Operational Risk Profile:** Heavy payloads (>6,000 kg) directed to GTO orbits represent high-risk missions where booster re-entry energy prevents recovery. In these flight profiles, competitors can compete on price because SpaceX is likely to expend the booster.
• **Launch Facility Advantage:** KSC LC-39A demonstrates operational maturity (76.92% success), whereas CCAFS SLC-40 exhibits higher landing volatility (42.86%).

Actionable Business Recommendations:

• **Competitive Bidding Strategy:** When competing for LEO or Starlink-like smallsat constellations (2,000 - 4,000 kg), expect SpaceX to bid aggressively at $62M due to a >72% landing probability. For GTO missions exceeding 6,000 kg, price bids assuming SpaceX must account for booster hardware loss.
• **Launch Site Contract Structuring:** Prioritize Florida-based orbital tracks where coastal recovery fleets are fully staffed and logistical supply chains (0.58 km to highways) minimize turnaround overhead.
• **Technological Roadmap:** Reusable heat shielding and grid-fin advancements modeled in Block 5 (100% success) are vital to achieving long-term economic parity.

Innovative Insights & Future Technical Roadmap

Financial Impact of Misclassifications, Class Rebalancing, and Enterprise Scalability

1. Financial Impact of Asymmetric Classification Errors (False Positives vs. False Negatives):

• **Zero False Negatives (FN = 0):** The model never predicted a crash when the booster actually survived. This ensures recovery fleet assets (drone ships, tugboats, recovery technicians) are never omitted when a booster lands.

• **False Positives (FP = 3):** In 3 instances, the model predicted a successful landing, but the booster was lost at sea. In a commercial setting, a False Positive causes dual losses: deployment costs of sea recovery vessels (~$200,000 to $500,000 per mission) plus unexpected disruption to scheduled re-flight payloads.

2. Innovative Remediation: Cost-Sensitive Loss Functions & SMOTE:

• **Class Imbalance Correction:** Because successful landings constitute 66.7% of the dataset, models naturally bias toward predicting success. Implementing **SMOTE** (Synthetic Minority Over-sampling Technique) or setting class_weight='balanced' will penalize False Positives and increase sensitivity to high-risk flight parameters.

3. Creativity & Presentation Architecture:

• Integrated a cohesive visual design system with high-contrast charts, custom geospatial distance visualizations, and structured data tables.

• Source Code & Artifacts: **https://github.com/alfaturachman/IBM-Applied-Data-Science-Capstone**